

Github link: <https://github.com/chenjie-gu/ds5500>

Weekly Update — NSF Funding Analysis

1. What did you accomplish this week?

Finished rebuilding the full analysis pipeline with cleaner topic labels after redoing the LDA model with a wider k sweep and selecting the best k based on coherence. Relabeled all topics manually to match the actual top words. Updated all subfield charts to use normalized 100% stacked bars so funding and award share can be compared year over year. Added distinct color schemes using Plotly qualitative palettes for better readability on the line charts. Also added PI-Institution hierarchical charts sorted by school, then professor, with a dropdown filter and exponential regression curves to the institution funding bar chart.

2. Did you learn or discover anything new?

The most surprising finding is that nearly every CS subfield is declining in both awards and funding from 2021 to 2025, except for a few areas like AI and computing infrastructure. STEM Education and Workforce had an anomalous funding spike in 2023, likely from CHIPS Act funding, then dropped sharply. The LDA and BERTopic models agreed well on clean topics like networking and cybersecurity, but diverged on interdisciplinary areas like health and education.

3. Are you uncertain, stuck, or worried about anything?

The LDA topic labels must be reassigned manually every time the model reruns, since topic numbering shifts so if the model gets rerun, the labels need to be re-verified against the printed topic words. Some smaller subfields have too few awards per year, so their growth percentages are volatile and may not be meaningful. Also, not 100% sure the 2025 data is complete since late-year awards may not be in the dataset, yet, which could be making the funding decline look worse than it is.

4. What is your plan for next week?

Write up the findings and interpretation, clean up the notebook with markdown explanations for each section, and finalize the PDF. On track overall.

5. Anything else?

No, the core analysis is complete.